

Paper III - Subspeciality of Urology (practice paper, MCh Urology Summer 2015 set)

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III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting.

Type: LAQ (25 marks)

Introduction / History

- A hollow tube keeping the upper tract draining past obstruction, injury or anastomosis.
- **Zimskind 1967**: first indwelling silicone splint; being straight, it migrated.
- **Finney 1978**: the **double-J (double-pigtail)** stent. Self-retaining coils prevent migration.

Stent design

Parameter	Details
Diameter	3–8.5 Fr (adults usually 6 Fr). Diameter does not affect symptoms
Length	Adults 20–30 cm by height (usually 24–26 cm); children about age + 10 cm

Diagram to draw: double-J stent in situ with renal coil, side holes and bladder coil.

Stent materials

Material	Features
Polymers (polyurethane, silicone)	Silicone is most biocompatible and least encrusting but soft; hydrocoated silicone causes less pain
Metallic (Resonance)	Tightly coiled alloy that resists extrinsic compression; malignant obstruction and retroperitoneal fibrosis; changed about yearly
Memokath 051	Thermo-expandable nitinol for long-term strictures
Biodegradable	No removal, so no forgotten stent; still experimental

Indications

- Relief of obstruction: obstructed infected kidney (stent and PCN (percutaneous nephrostomy) are equally effective), malignant obstruction, stricture, pregnancy.
- After intervention: selected URS (ureteroscopy), pyeloplasty, reimplantation, transplant.
- Healing of injury, leak or fistula; pre-stenting a tight ureter.

Stent-related symptoms

- About four in five patients have frequency, urgency, haematuria or flank pain on voiding, measured by the validated **USSQ (Ureteral Stent Symptom Questionnaire)**.
- Mechanism: **intravesical stent irritating the bladder** and **reflux of urine up the stent** during voiding.
- Relief: **alpha-blockers and antimuscarinics**, less intravesical material, shortest dwell time.

Complications

- **Misplacement** (submucosal, below the block) and **migration** (stent too short).
- **Stent failure** from compression, debris or encrustation; highest with malignant or irradiated ureters. New obstructive symptoms need imaging.
- **Encrustation** rises with dwell time, infection and stone-forming urine; it risks fracture or avulsion on removal.
- **Forgotten stent**: obstruction, sepsis, renal loss; a stent register prevents it.
- **FECal (Forgotten, Encrusted, Calcified) grading**: I minimal linear; II circular; III partial encrustation of one or both coils; IV complete encrustation of one coil; V both coils.

Management of an encrusted stent

- Never pull by force. Assess with NCCT (non-contrast computed tomography); treat infection first.
- Clear the **bladder coil** first (cystolithotripsy), then the **ureter** (URS and laser), then the **renal coil** (flexible URS, or PCNL (percutaneous nephrolithotomy) for a large burden).

Ureteroscopy

	Semirigid	Flexible
Tip	7 Fr or less	4.5–7.5 Fr; 3.6 Fr working channel
Best for	Distal and mid ureter	Upper ureter and kidney
Deflection	None	More than 270°, both directions

- **Safety guidewire first**: it holds access and allows a stent after perforation.
- **UAS (ureteral access sheath)**, usually 12/14 Fr, lowers intrarenal pressure and eases repeated passes, but caused **severe ureteric injury in about one in eight**; pre-stenting lowers this.
- **Ho:YAG (holmium:yttrium-aluminium-garnet) laser** is standard; **TFL (thulium fibre laser)** the alternative. Dusting uses low energy at high frequency.
- A **tight ureter** is not forced: stent and return after **2–4 weeks**.
- Complications: sepsis about 5%, stricture about 3%, avulsion under 1%.

Stenting after URS

- **Denstedt (2001)** randomised trial and a Cochrane review: URS is safe without a stent.
- **AUA (American Urological Association)**: the stent may be omitted when **all four** hold: normal anatomy, normal renal function, no ureteric injury, no planned second stage. **EAU (European Association of Urology)** also advises against routine stenting.
- **Stent required**: perforation, stricture or tight ureter, infection, planned second stage; also solitary kidney, impacted stone, pregnancy.
- Removal at 3 days gave more unplanned returns than at 7 days. A string allows home removal.

End of URS -> uncomplicated? (normal anatomy and function, no injury, no 2nd stage)
Yes -> no stent (or short string stent)
No -> double-J: injury 2-6 wk; residual/impacted or 2nd stage until re-look

Recent advances

- Suction access sheaths, single-use digital ureteroscopes, intrarenal pressure sensors, biodegradable stents.

Key points to remember:

- Finney 1978: double-J stent.
 - Symptoms come from intravesical stent and reflux; alpha-blocker plus antimuscarinic helps.
 - Encrusted stent: clear bladder coil, then ureter, then kidney.
 - Denstedt 2001: no routine stent after uncomplicated URS (AUA four conditions).
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III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR.

Type: LAQ (25 marks)

Definition

- **VUR (vesicoureteral reflux):** retrograde flow of urine from bladder to upper tract.
- **Primary VUR:** congenital deficiency of the UVJ (ureterovesical junction) flap-valve; tunnel length to ureteric diameter should be about **5:1** (Paquin).
- **Secondary VUR:** high bladder pressure, e.g. PUV (posterior urethral valves), neurogenic bladder, BBD (bladder and bowel dysfunction).
- Found in about **30% of children with UTI (urinary tract infection)** and about 30% of siblings.

Pathophysiology

- **Infection, not reflux, is the injury.** Reflux carries infected urine to the kidney; compound papillae allow intrarenal reflux and **scarring**.
- Scarring causes hypertension, proteinuria and CKD (chronic kidney disease): **reflux nephropathy**.

International (IRS, International Reflux Study) grading on VCUG (voiding cystourethrogram)

Grade	Findings
I	Non-dilated ureter only
II	Ureter, pelvis and calyces, no dilatation
III	Mild-moderate dilatation, minimal forniceal blunting
IV	Moderate dilatation and tortuosity; papillary impressions maintained
V	Gross dilatation and tortuosity; papillary impressions lost

Diagram to draw: grades I-V sketches and a Cohen cross-trigonal reimplant.

Natural history

- Resolution falls with grade and age: grade II 63–85%, grade III about 50%, grade IV–V rarely.
- **Bilateral grade IV persists most** (AUA (American Urological Association) 1997: about 10% resolve in 5 years).

Evaluation

- History of febrile UTI and BBD; BP (blood pressure), spine, urine culture, creatinine, urine protein.
- **RBUS (renal-bladder ultrasound):** renal size, hydronephrosis, residual. A normal scan does not exclude VUR.
- **VCUG:** grade; urethra in boys (PUV).
- **DMSA (dimercaptosuccinic acid) scan:** scars and differential function.

Medical management

1. Principle: prevent UTI and scars while awaiting resolution. High risk: female, bilateral, grade III–V, BBD, scars, multiple febrile UTIs.

2. CAP (continuous antibiotic prophylaxis): single nightly dose, about one-quarter of the treatment dose.

Drug	Dose	Notes
TMP-SMX (trimethoprim-sulfamethoxazole)	2 mg/kg TMP	Avoid under 2 months (kernicterus)
Nitrofurantoin	1–2 mg/kg	Avoid in G6PD (glucose-6-phosphate dehydrogenase) deficiency
Trimethoprim, amoxicillin or a first-generation cephalosporin	—	Young infants

3. Evidence

- **RIVUR (2014):** 607 children, grade I–IV, TMP-SMX vs placebo. Recurrent UTI fell from **23.6% to 12.9%**, but **scarring was unchanged**. Benefit was greatest with BBD; only about 8% had grade IV.
- **Swedish Reflux Trial:** age 1–2 years, grade III–IV. In **girls**, febrile UTI recurred in 19% on CAP, 23% after endoscopy and 57% on surveillance. **Boys showed no difference.**

4. Treat BBD first: timed voiding, treat constipation, anticholinergics. BBD increases breakthrough UTI and lowers resolution and endoscopic success.

5. Circumcision: in boys with VUR, about 4 circumcisions prevent one UTI.

Indications for intervention: breakthrough febrile UTI on CAP, new scars, persistent high-grade VUR, non-compliance.

Case: 8-year-old boy with bilateral grade IV VUR

Assessment

- Exclude PUV, neurogenic bladder and BBD. DMSA, creatinine, BP, UTI history.
- At 8 years bilateral grade IV rarely resolves, but older boys have a low UTI risk, so treating the asymptomatic boy is **controversial**.

Options

1. **CAP with BBD treatment:** if no febrile UTI, no scars, normal bladder.
2. **Endoscopic injection (Dx/HA, dextranomer/hyaluronic acid, Deflux)** by STING (subureteric Teflon injection technique) or HIT (hydrodistension implantation technique). Lower success in grade IV (about 60% per injection); 20% recurred by 2 years in the Swedish trial.
3. **Ureteral reimplantation:** open success **95.9%** (Elder meta-analysis 1997).
 - **Cohen cross-trigonal** (intravesical): commonest, over 98% success; one bladder opening corrects both sides; later ureteroscopy is difficult.
 - **Lich-Gregoir** (extravesical): no haematuria or spasm, but **transient retention in up to 10% after bilateral repair** (pelvic plexus injury).
 - **RALUR (robot-assisted laparoscopic ureteral reimplantation):** less pain; results approaching open.

Recommended plan

- Breakthrough febrile UTI, scars or falling function: **bilateral Cohen reimplantation** (Deflux is less durable).
- Asymptomatic, normal kidneys: CAP or observation with BBD therapy; correct if febrile UTI or new scar.

- After surgery: ultrasound at 6–12 weeks; VCUG after high-grade repair; long-term BP checks.

Exclude PUV / neurogenic bladder, treat BBD

-> febrile UTI, scars or renal loss? Yes -> bilateral Cohen (or Deflux)

No -> CAP + BBD therapy, correct if UTI/scar

Recent advances

- ceVUS (contrast-enhanced voiding urosonography); UDR (ureteral diameter ratio) predicts resolution better than grade.

Key points to remember:

- IRS grade IV keeps papillary impressions; grade V loses them.
- RIVUR 2014: CAP halves recurrent UTI (23.6% to 12.9%) but does not change scarring.
- Open reimplantation 95.9%; Cohen over 98%; Deflux lower in grade IV.
- Bilateral extravesical repair risks transient retention (up to 10%).

III-3a. Sacral neuromodulation

Type: SAQ (10 marks)

Definition and mechanism

- **SNM (sacral neuromodulation):** continuous low-amplitude **S3 root** stimulation via a tined lead and an IPG (implantable pulse generator).
- It acts through **afferents, not motor nerves**, on forebrain areas of awareness. In OAB (overactive bladder) this damps the PMC (pontine micturition centre); in retention it restores filling sensation, so the PMC switches on and the sphincter relaxes. Hence it treats **both overactivity and retention**.

Indications

FDA (Food and Drug Administration) approved	Not approved
Refractory UUI (urgency urinary incontinence), 1997	Bladder pain syndrome
Urgency-frequency and non-obstructive retention (e.g. Fowler's syndrome), 1999	Neurogenic bladder
Faecal incontinence, 2011	Pelvic pain

- Third-line OAB therapy in American and European guidelines.

Technique

- **Test phase: PNE (percutaneous nerve evaluation)** for **5-7 days**, or a **staged tined lead** for up to 4 weeks (more reliable; preferred for retention).
- S3 foramen: about **9 cm above the coccyx tip, 1-2 cm lateral**.
- **S3 response:** perineal "bellows" plus **great-toe flexion**.
- **Pass:** at least **50% improvement** in a diary parameter; then implant the IPG in the buttock.

Outcomes

- **InSite (2015):** success 61% with SNM vs 42% with medical therapy at 6 months.
- **ROSETTA (2016):** onabotulinumtoxinA 200 U reduced UUI slightly more, but caused more UTIs (urinary tract infections) (24% vs 10%) and self-catheterisation.

Complications

- IPG-site pain, lead migration, loss of effect, infection; **revision over 30% at 5 years**.

Contraindications

- Failed test, inability to use the programmer.

Recent advances

- Rechargeable devices (15 years or more); **full-body MRI (magnetic resonance imaging) conditional at 1.5 and 3 T**.

Diagram to draw: sacrum with the lead in S3 and the IPG in the buttock.

Key points to remember:

- Afferent modulation; treats both overactivity and retention.
- S3 response: bellows plus great-toe flexion.
- Implant if the test gives at least 50% improvement.

III-3b. Uses of buccal mucosa in urology

Type: SAQ (10 marks)

Introduction / history

- **BMG (buccal mucosal graft)** is the workhorse oral mucosal graft for urethral reconstruction.
- **Humby 1941** first used it for hypospadias; **Bürger 1992** and **El-Kasaby 1993** popularised it for stricture.

Advantages

- Thick, **non-keratinised**, hairless epithelium used to a wet environment.
- **Thin, highly vascular lamina propria** gives rapid imbibition and inosculation, so take is excellent.
- Minimal contraction, easy harvest, hidden scar, low donor morbidity.

Harvest technique

- Nasotracheal intubation. Graft about 4–6 × 2–2.5 cm per cheek.
- Stay about **1 cm from the lip commissure** and **avoid Stensen's (parotid) duct** opposite the upper second molar.
- Excise mucosa without muscle, **defat** the graft, cut it about 10% larger than the defect.
- Alternatives: **lingual** and **lip** grafts (lip is thinner, better in the glans).

Donor-site complications

- Pain, numbness, mouth tightness, duct injury; mostly transient.

Uses

Area	Use
Bulbar urethroplasty (main use)	Dorsal onlay (Barbagli), ventral onlay, Asopa dorsal inlay, augmented anastomotic repair
Panurethral stricture	Kulkarni one-stage dorsal onlay via a perineal incision
Staged urethroplasty (Johanson, Bracka)	Graft first; tubularise after at least 6 months. For lichen sclerosis and failed hypospadias
Hypospadias	Redo cases; dorsal inlay when the plate is narrow
Others	BMG perineal urethrostomy, robotic ureteroplasty, bladder neck contracture, female urethral stricture

Outcomes

- Bulbar BMG urethroplasty succeeds in about 85–90%. Dorsal equals ventral onlay; graft equals flap.
- Results fall with time and in penile, lichen sclerosis and redo cases. **Avoid skin in lichen sclerosis.**

Recent advances

- Tissue-engineered oral mucosa (investigational).

Diagram to draw: cross-section of the bulbar urethra showing dorsal onlay, ventral onlay and Asopa inlay.

Key points to remember:

- Thin vascular lamina propria gives excellent take; avoid Stensen's duct.
 - Staged BMG (Johanson/Bracka) for lichen sclerosus.
 - Bulbar BMG urethroplasty success about 85-90%.
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III-3c. HSG in neonates

Type: SAQ (10 marks)

HSG (hysterosalpingography) is not done in neonates, so this is taken as "USG (ultrasonography) in neonates".

Introduction

- First-line, radiation-free imaging for antenatal **UTD (urinary tract dilation)** (1-3% of pregnancies), febrile UTI (urinary tract infection) and masses.

Timing

- **First scan after 48 hours:** newborns are relatively dehydrated, so earlier scans under-read dilatation.
- **Scan early if outlet obstruction is suspected** (bilateral severe dilatation, thick bladder); catheterise and start prophylaxis.

Normal neonatal kidney

- **Prominent hypoechoic pyramids** are normal, not hydronephrosis.
- Measure pelvic **APD (anteroposterior diameter)** on a transverse mid-kidney view; normal postnatal APD is **under 10 mm**.

Grading

- **SFU (Society for Fetal Urology) 0-4:** 1 pelvis only; 2 major calyces; 3 minor calyces; 4 plus parenchymal thinning.

UTD classification 2014, postnatal (worst feature decides)

Grade	Findings	Management
P1 (low)	APD 10 to <15 mm or central calyces	USG at 3-6 months; no VCUG (voiding cystourethrogram) or prophylaxis
P2 (intermediate)	APD 15 mm or more , peripheral calyces or dilated ureter	VCUG and prophylaxis at discretion; prophylaxis if ureter >7 mm
P3 (high)	Abnormal parenchyma or bladder , any APD	USG at 1 month, VCUG, prophylaxis; MAG3 (mercaptoacetyltriglycine) renogram if persistent

Findings in specific conditions

Condition	USG findings
PUJO (pelviureteric junction obstruction)	Dilated pelvis, normal ureter
VUR (vesicoureteral reflux)	Up to 25% with VUR have a normal first scan
PUV (posterior urethral valves)	Bilateral hydroureteronephrosis, thick-walled bladder , dilated posterior urethra (" keyhole sign ", highly suggestive of PUV); confirm with VCUG
MCDK (multicystic dysplastic kidney)	Non-communicating cysts, scant parenchyma

Recent advances

- ceVUS (contrast-enhanced voiding urosonography); pyramid thickness under 3 mm predicts pyeloplasty.

Diagram to draw: SFU grades 0–4.

Key points to remember:

- First scan after 48 hours, except suspected PUV or bilateral severe dilatation.
 - UTD 2014: P1 APD 10 to <15 mm; P2 15 mm or more; P3 abnormal parenchyma or bladder.
 - Keyhole sign (thick bladder plus dilated posterior urethra) suggests PUV.
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III-3d. Types of penile prosthesis. What are biofilms in penile prosthesis?

Type: SAQ (10 marks)

Introduction

- **Scott (1972)** inflatable; **Small-Carrion (1978)** semirigid. For ED (erectile dysfunction) failing less invasive treatment; satisfaction **80-90%**.

Types

Type	Examples	Advantages	Disadvantages
Malleable (semirigid)	Tactra (nitinol), Genesis (silver)	Simple, cheap, no mechanical failure	Always rigid, hard to conceal
2-piece inflatable	Ambicor (now discontinued)	No reservoir	Poor flaccidity
3-piece inflatable	AMS 700 CX/LGX, Coloplast Titan	Best rigidity with true flaccidity	Cost, mechanical failure

- 3-piece: paired cylinders, scrotal pump, reservoir in the space of Retzius or an **ectopic submuscular** site after pelvic surgery. Repairs needed in about **15% at 5 years and 30% at 10 years**.

Biofilm

- **Definition:** a structured community of microorganisms in a self-produced **polysaccharide matrix**, attached to the implant.
- **Stages:** attachment, microcolonies, maturation, dispersal (clinical infection).
- **Organisms:** coagulase-negative staphylococci (*S. epidermidis*) commonest; also *S. aureus*, Gram-negatives, *Candida*.
- **Why antibiotics fail:** poor matrix penetration and slow-growing bacteria, which survive up to **1000-1500 times** the usual killing concentration.
- **Relevance:** organisms from the first operation lie dormant. Revising an uninfected device carries about 10% infection; an **antiseptic washout** lowers it to about 3%.

Prevention

- **Infection 1-3%**; higher with diabetes and revision.
- **Chlorhexidine-alcohol** skin preparation; intravenous antibiotics (vancomycin or cephalosporin plus gentamicin) for up to 24 hours.
- **Coated devices halve infection:** **InhibiZone** (rifampin-minocycline) on AMS; hydrophilic coating soaked in antibiotic on Coloplast.
- **No-touch technique (Eid):** under 0.5% infection.

Management of infection

- Remove **all** components.
- **Mulcahy salvage:** explant, antiseptic washes (dilute povidone-iodine, then saline), immediate new device. Success **84%**; fails with early onset, aggressive organisms or pus with cellulitis.
- Otherwise delayed reimplantation (loses length).

Diagram to draw: 3-piece device in situ and the biofilm life cycle.

Key points to remember:

- Types: malleable, 2-piece and 3-piece inflatable (best rigidity and flaccidity).
 - Biofilm is a microbial community in a matrix; *S. epidermidis* is commonest.
 - Infection 1-3%; coatings halve it; Mulcahy salvage succeeds in 84%.
-

III-3e. Uretero-vaginal fistula

Type: SAQ (10 marks)

Definition

- **UVF (uretero-vaginal fistula):** a ureter-vagina communication causing continuous leakage.

Aetiology

- Usually **unrecognised injury to the distal ureter**; about **three-quarters follow abdominal hysterectomy**, often uneventful. The ureter is at risk **under the uterine artery** and at the vaginal fornix.
- Others: obstetric injury, cancer, radiation.

Clinical features

- **Continuous leakage with normal voiding**, because the bladder still fills from the other kidney.
- Appears **within 4 weeks**, often with fever or flank pain, as thermal or ischaemic injury necroses.

Investigations

- **Double dye test:** oral phenazopyridine (orange urine) plus blue dye in the bladder. **Orange tampon = UVF**, blue = VVF (vesicovaginal fistula), both = both. Three-swab test: a wet unstained swab suggests UVF.
- **CT (computed tomography) urography, delayed phase:** leak and hydroureteronephrosis.
- **Cystoscopy and retrograde pyelogram:** site; stent at the same sitting.
- **Exclude coexisting VVF** (up to 25% of VVF have a UVF) and bilateral injury.

Management

1. Stent first

- **Retrograde double-J** if a wire crosses; otherwise PCN (percutaneous nephrostomy) and antegrade stent.
- **Stenting within 2 weeks** of detection works best. Remove at **4-6 weeks** and re-image; most heal.

2. Surgical repair (stent fails)

- **Ureteroneocystostomy** with **psoas hitch** (about 5 cm) or **Boari flap** (up to 15 cm).
- Early repair succeeds as well as delayed (no difference before or after 6 weeks); success **90% or more**.

Wet + normal voiding -> double dye, CT urography, cystoscopy (exclude VVF)
-> wire passes? Yes -> double-J 4-6 wk; No -> PCN + antegrade stent
-> fails -> ureteroneocystostomy + psoas hitch / Boari

Follow-up and prevention

- Ultrasound after stent removal; stricture may appear up to 30 months later.
- Prevention: identify the ureter; prophylactic stents only if high risk.

Diagram to draw: double dye test; psoas hitch.

Key points to remember:

- Continuous leak with normal voiding; orange tampon on double dye test = UVF.
 - Exclude coexisting VVF (up to 25% of VVF have UVF).
 - Early stent (within 2 weeks); reimplant with psoas hitch if it fails.
-

III-3f. Prune belly syndrome

Type: SAQ (10 marks)

Definition

- **PBS (prune belly syndrome)**, Eagle-Barrett or triad syndrome:
 1. Deficient abdominal wall muscles (wrinkled "prune" abdomen).
 2. Dilated, poorly contracting urinary tract.
 3. Bilateral intra-abdominal testes.
- About **1 in 29,000-44,000** live births; **95% male. Renal dysplasia, not the abdominal wall, decides prognosis.**

Pathogenesis (unproven)

- **Obstruction theory:** transient fetal outlet obstruction, though most have none after birth.
- **Mesodermal theory:** lateral plate mesoderm defect affecting abdominal wall, urinary tract and gubernaculum. Mostly sporadic.

Genitourinary features

- Kidneys: dysplasia in about 50%.
- Ureters: dilated, tortuous, **worse distally**; collagen replaces muscle, so peristalsis fails. **VUR (vesicoureteral reflux) about 75%.**
- Bladder: large, smooth, urachal diverticulum. Dilated prostatic urethra; infertility.

Extra-genitourinary (about 75%)

- Pulmonary hypoplasia; cardiac about 10%; gastrointestinal (malrotation) at least 30%; orthopaedic up to 65%.

Woodard classification

Category	Features	Outlook
I	Oligohydramnios, pulmonary hypoplasia , renal dysplasia	Perinatal death; supportive care
II	Full triad, moderate uropathy, no pulmonary hypoplasia	Risk of sepsis and CKD (chronic kidney disease)
III	Mild or incomplete, stable renal function	Good; the majority

Management

- **Neonatal:** cardiorespiratory care first; creatinine trend, ultrasound, prophylaxis, circumcision. VCUG (voiding cystourethrogram) only if renal insufficiency or obstruction.
- **Category III:** conservative.
- **Drainage** for sepsis, rising creatinine or obstruction: wide **cutaneous vesicostomy**.
- **Reconstruction** after 3-6 months: ureteric tailoring and reimplantation using the healthier proximal ureter.
- **Orchidopexy:** bilateral **transabdominal at about 6 months**; staged Fowler-Stephens if vessels are short.
- **Abdominoplasty:** Randolph, Ehrlich or **Monfort**, often with orchidopexy.

Prognosis

- About 20% die perinatally; **up to 30% develop chronic renal failure**. Nadir creatinine **under 0.7 mg/dL** predicts stable function. Transplant results are good.

Diagram to draw: VCUG with large bladder, urachal diverticulum and dilated prostatic urethra.

Key points to remember:

- Triad: deficient abdominal wall, dilated urinary tract, intra-abdominal testes; 95% male.
 - Renal dysplasia decides prognosis; Woodard I-III.
 - Vesicostomy if drainage needed; orchidopexy at about 6 months; nadir creatinine under 0.7 mg/dL is good.
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Rapid Revision Sheet - Paper III

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

III-1. Discuss various ureteric stents. Discuss ureteroscopy with or without stenting. (LAQ)

- Finney 1978: double-J stent.
- Symptoms come from intravesical stent and reflux; alpha-blocker plus antimuscarinic helps.
- Encrusted stent: clear bladder coil, then ureter, then kidney.
- Denstedt 2001: no routine stent after uncomplicated URS (AUA four conditions).

III-2. Describe medical management of VUR. Describe treatment of an 8-yr male child with b/l grade IV VUR. (LAQ)

- IRS grade IV keeps papillary impressions; grade V loses them.
- RIVUR 2014: CAP halves recurrent UTI (23.6% to 12.9%) but does not change scarring.
- Open reimplantation 95.9%; Cohen over 98%; Deflux lower in grade IV.
- Bilateral extravesical repair risks transient retention (up to 10%).

III-3a. Sacral neuromodulation (SAQ)

- Afferent modulation; treats both overactivity and retention.
- S3 response: bellows plus great-toe flexion.
- Implant if the test gives at least 50% improvement.

III-3b. Uses of buccal mucosa in urology (SAQ)

- Thin vascular lamina propria gives excellent take; avoid Stensen's duct.
- Staged BMG (Johanson/Bracka) for lichen sclerosus.
- Bulbar BMG urethroplasty success about 85–90%.

III-3c. HSG in neonates (SAQ)

- First scan after 48 hours, except suspected PUV or bilateral severe dilatation.
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- Types: malleable, 2-piece and 3-piece inflatable (best rigidity and flaccidity).
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- Continuous leak with normal voiding; orange tampon on double dye test = UVF.
- Exclude coexisting VVF (up to 25% of VVF have UVF).
- Early stent (within 2 weeks); reimplant with psoas hitch if it fails.

III-3f. Prune belly syndrome (SAQ)

- Triad: deficient abdominal wall, dilated urinary tract, intra-abdominal testes; 95% male.
- Renal dysplasia decides prognosis; Woodard I–III.
- Vesicostomy if drainage needed; orchidopexy at about 6 months; nadir creatinine under 0.7 mg/dL is good.